AI BUILDER CUP 2026 · MANUFACTURING

OSI

Opsight Intelligence

At 2 a.m., two machines need the last spare part. OSI decides who gets it, covers the other, and proves it paid off.

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THE PROBLEM

Predicting the failure is the easy part. Getting the part in time is not.

A model can say a spindle fails in a day. Whether the line keeps running depends on three questions in three other systems.

Stores

Is the bearing on the shelf, or already promised to another machine?

Purchasing

When will the open order really arrive, given how this supplier delivers?

Customer service

Which customer orders does this line make in the next few days?

Today the answer takes hours of phone calls, while the warning window (the P-F interval) runs out.

The expensive mistakes happen between departments.

Predictive maintenance stops at the alert. OSI carries the alert across stores, purchasing, and customer service, and turns it into a plan a person approves.

THE SOLUTION

Sense, reason, act, and close the loop

1 Sense

A gradient-boosting model scores every machine from its sensors. It caught 27 of 41 failures, with a median warning of 86 hours.

2 Reason

A team of Gemini agents checks sensors, photos, spares at both sites, the supplier's record, and customer orders.

3 Act

The team drafts the work order, the parts action, and the customer note. A person approves.

4 Close the loop

Approval moves stock and clears conflicts. Once the failure date passes, the outcome is scored.

In live mode the plant replays in time. With Autopilot on, each new alert pauses the clock while the agents work, so the plan is ready by morning.

HOW A REAL PLANT CONNECTS

Your systems feed one governed layer

The plant's system	What OSI takes from it	What the agents can answer
Sensor historian	Sensor history, live readings	Which machine fails, and when
Maintenance system (CMMS)	Work orders and repairs	What fixed it last time
ERP: stores, purchasing	Stock, open orders, deliveries	Is the part here, or when it arrives
ERP: order book	Customer orders per line	Which customers are at risk
Inbox and shop floor	Supplier mail, inspection photos	Did the date move; is the wear real

Mappings, not prompts

The agents only see governed tools. Connecting a plant means mapping its data, not rewriting the AI.

One file per site

The governed bridge links each site to its supply plant, its parts to spares and suppliers, its lines to customers.

ONE ALERT, START TO FINISH

25 July 2026, 02:00: two alerts and one bearing set on the shelf

25 Jul 02:00

Two spindle-bearing alerts at site US1 need the one bearing set on the shelf.

Clock paused

The supplier was on time for 66.1% of lines. Gemini reads its delay notice: new date 8 August.

Plan

Air transfer from site DE1 for $650. Waiting would mean 190 h down and 4 customer lines at risk.

26 Jul 10:00

A person approved. The spindle fails as predicted, and the part is already there.

26 h

down with the plan, instead of 230 h

$100,077

saved on this one alert

PROBLEM ALIGNMENT AND IMPACT · 25%

Every alert of four months, scored against what really happened

$197,949
net saved, after charging the plan for false alerts

662 h
of downtime avoided

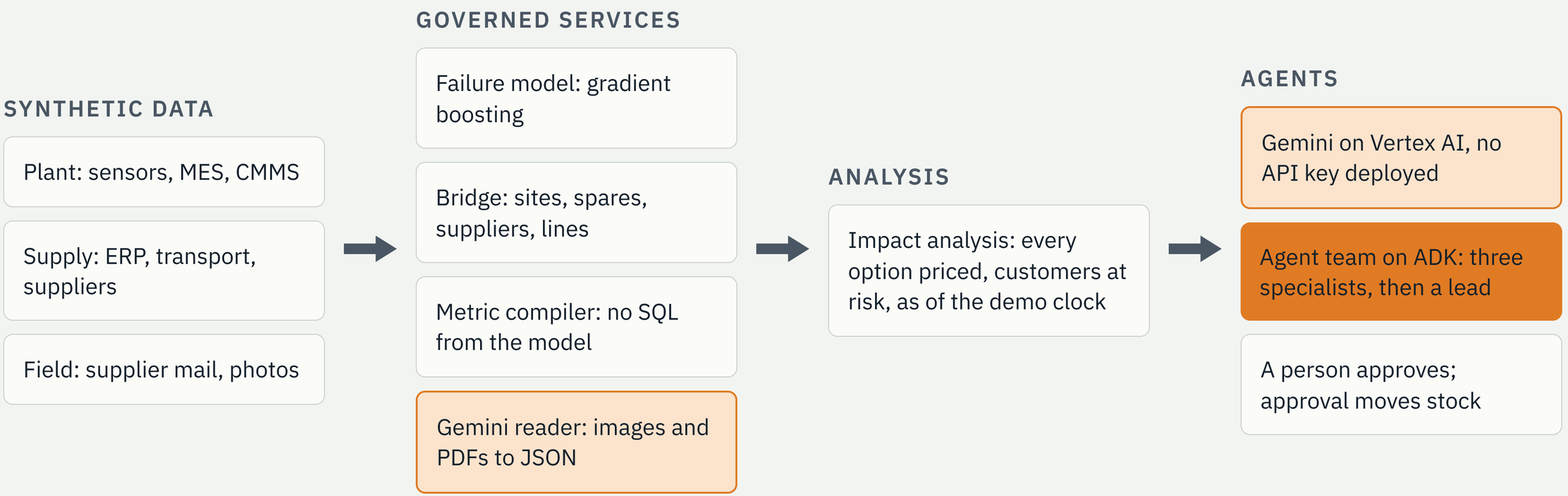
38
alerts scored: 26 real failures, 12 false

Plans that differed	Saved on real failures	Spent on false alerts	Net
6 of 38	$199,809	$1,860	$197,949

Each alert compares OSI's plan with what the site would have done alone. Reproduce it with make value.

TECHNICAL MERIT AND GEN AI · 40%

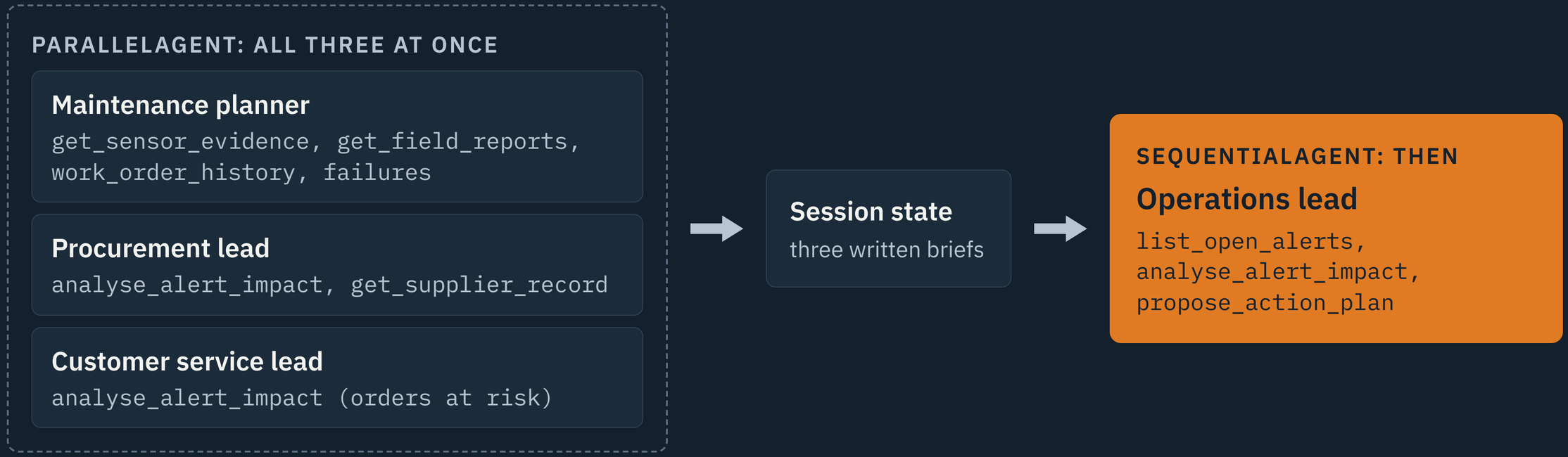
One Cloud Run service, Gemini on Vertex AI



Orange parts use Gemini. Prediction, pricing, and metrics are ordinary code, so every number an agent quotes is computed, then checked. Gemini never decides whether an alert is real: the model raises it, sensors and photos test it, a person decides.

TECHNICAL MERIT AND GEN AI · 40%

An agent team on Google's ADK



Each specialist sees only its department's tools. The lead weighs urgency, cost, and customer exposure, then proposes. Backup Gemini models take over when one runs out of quota.

TECHNICAL MERIT AND GEN AI · 40%

Gemini reads supplier mail and photos from the floor

Supplier inbox

Text or PDF in, a schema-checked delivery update out. A delay notice moves the open order to its new date in every later analysis.

Inspection photos

Gemini vision reads a technician's photo against the predicted failure mode, before any money is spent.



US1-L2-01 bearing: confirmed, 0.95



DE1-L2-04 gearbox: not confirmed, 0.88

The gearbox alert was false in the data, and Gemini saw intact teeth. Five photos tested: three worn parts confirmed, a photo of the wrong part flagged.

TECHNICAL MERIT AND GEN AI · 40%

Why the numbers can be trusted

No invented numbers

Every number in an answer is checked against the tool results before it is marked verified.

No SQL from the model

Supplier metrics compile from governed definitions and carry a SQL fingerprint.

No peeking at the future

Every tool answers as of the demo clock. Outcomes count only once they are in the past.

No silent actions

Agents only propose, with numbers from the analysis, not the model's text. People approve.

Evidence: 423 backend and 7 frontend tests, offline tests of the real ADK runner, and an eight-scenario agent evaluation: **8 of 8 with Gemini on Vertex AI**.

INNOVATION AND CREATIVITY · 25%

What is new here

Agents organised like the plant	Maintenance, procurement, and customer service specialists, each with only its own tools, and a lead who weighs their briefs.
An autopilot for the night shift	Live mode pauses the clock on each new alert and leaves a plan ready by morning.
Approvals change the world	Decisions move stock and clear conflicts; outcomes are scored once the clock passes the failure.
Vision as a check before spending	A photo from the floor confirms or rejects the prediction before an express buy.
An honest track record	False alerts are charged against the plan, not hidden.
Operators teach it, never silence it	A false alert marked by a person makes the next one ask for an inspection, not disappear. Two machines failed weeks after a false alert.

USER EXPERIENCE AND DESIGN · 10%

Built for the people on shift

Page	What it shows
Live	The plant replayed in time, with the autopilot
Control room	Alerts, spare conflicts, priced options, sensor evidence, customers at risk
Copilot	Questions to one agent or to the team
Supplier inbox	Supplier messages read into delivery updates
Approvals	Approve or reject; outcomes appear later
Track record	Every alert scored at its real outcome
How it works	The loop, the architecture, the guardrails

Watch the team work

Every tool call and brief streams live with the agent's name.

A verified badge

Shown only when every number matches the tools.

Errors in plain words

Quota, overload, or credentials, each with what to do.

SCALABILITY

From prototype to a real plant on Google Cloud

Part	Prototype today	At scale
Operational state	JSON files in the container	Firestore
Plant and supply data	DuckDB over Parquet, from seeds	BigQuery, fed from MES, CMMS, and ERP
Sensor and event stream	Replayed from history	Pub/Sub
Failure model	Trained at build time	Vertex AI training and endpoints
Monitoring	Container logs	Cloud Monitoring for agent latency and errors
More sites	Two sites, four supply plants	More mappings in the governed bridge

Cloud Run scales to zero when idle. A team run costs about twelve Gemini Flash calls. Sites, spares, suppliers, and lines are mapped in governed YAML, so a new site starts with data mappings.

HONEST LIMITS

What this prototype does not claim

- The data is synthetic: a two-site plant and a four-plant supply network generated from seeds. No real company data.
- Spare costs and contribution per unit are declared assumptions in the governed bridge file.
- The track record scores each alert independently.
- Approvals and field inputs reset when the container restarts, until they move to Firestore.

From an alert at 2 a.m. to an approved plan before anyone arrives.

Live app ops-copilot-432875717420.asia-southeast1.run.app

Video youtu.be/b1bc3dhXC74

Code github.com/UttyWotty/ai_build_base_2026

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